MANOJ PAWAR SJ

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SUMMARY

Backend engineer (SDE 2) with nearly 5 years of experience building distributed, real-time and event-driven systems in Go, Python, TypeScript and Node.js across space tech, e-commerce, gaming and CRM. Currently onsite at ISTRAC, ISRO, building a real-time monitoring and control platform. Cut cloud costs by **40%** with concurrent Go services, made a batch pipeline **6x faster** and removed **90%** of manual ops work with a self-serve scheduler; earlier, lifted e-commerce conversion by **5%** and CRM call pickup rates by **35%**.

TECHNICAL SKILLS

Languages: Go (Golang), Python, TypeScript, JavaScript, SQL
Backend: Node.js, Nest.js, Express.js, Hono, REST APIs, Microservices, WebSockets, SSE, MQTT, TCP/IP sockets
Distributed Systems: Event-driven architecture, Pub/Sub, Real-time streaming, Rules engines, Binary protocol parsing, RBAC, Caching (Redis)
Concurrency: Go goroutines & channels, Python asyncio, threading, multiprocessing
Databases & ORMs: PostgreSQL, MySQL, MongoDB, Redis, Firestore, TypeORM, Drizzle, Sequelize, Mongoose
Cloud & DevOps: AWS (ECS, EC2, Lambda, SQS, S3, RDS, Route 53), GCP, Docker, Terraform, Gitea CI/CD, Git, Linux (RHEL 9)
Observability: Grafana, Loki, OpenObserve (log storage and querying)
Testing: pytest, Vitest
AI Tooling: Claude Code for design, coding, code review and debugging; llama.cpp to run local models offline, with OpenCode for agentic work on offline systems
Frontend: React.js, Next.js, Tailwind CSS
Other: SciPy, SGP4, Azure PlayFab, Firebase, Metabase

PROFESSIONAL EXPERIENCE

Digantara Industries | digantara.co.in | Bangalore, India | 10/2023 – Present
Software Development Engineer 2 | 03/2026 – Present
Onsite contract engineer at ISTRAC, ISRO | 02/2026 – Present

- Building backend services in Python and Node.js for a distributed, real-time **monitoring and control platform** across ISRO's ground station network, developed on an **offline, air-gapped RHEL 9 (Linux)** network under ISRO security rules.
- Integrated heterogeneous station hardware over raw **TCP/IP and Telnet sockets**, decoding **binary** and ASCII data streams with bit-level parsing; new device types are added through **configuration alone**, with no code changes.
- Designed a **config-driven rules engine** that evaluates live data for alarm detection, so operators change conditions with **no programming knowledge**; worked directly with ISRO operators to define alarm rules and workflows.
- Built real-time APIs with **SSE** streaming, user management and **RBAC**, and bandwidth-optimized **MQTT** pub/sub that sends only changed data.

Software Engineer | 10/2023 – 02/2026

- Optimized Go APIs that predict satellite positions and check for possible collisions, using **concurrent processing with goroutines and channels** and low memory use, **cutting cloud costs by 40%**.
- **Owned design and delivery** of the backends for 2 commercial products, **FDS** (satellite orbit and manoeuvre planning) and **SPACEMAP** (space object tracking), from requirements to production: ran design reviews, estimated story points, distributed tasks across the team and led weekly stand-ups; set up **Gitea CI/CD** pipelines deploying both to **AWS ECS**.
- **Reviewed code and mentored** engineers on the team.
- Built a self-serve **job scheduler in Nest.js** with role-based access control and user signup, letting teams change job timing without redeploys and **cutting manual ops work by 90%**.
- Rebuilt the batch collision-screening pipeline, cutting each run from **6 hours to 1 hour (6x faster)** through bulk precomputation, early candidate filtering and an optimized search (Python, SGP4, SciPy).
- Built the FDS web app in React.js and Tailwind CSS, with a focus on responsiveness and ease of use.

Elite Competitions | elitecompetitions.co.uk | Remote | 02/2023 – 07/2023
Software Engineer (Full Stack, Contract)

- Built event-driven communication between services using Nest.js, Next.js, **AWS Lambda and SQS**.
- Built a coupon system (one-time, user-specific and newsletter codes) that raised checkout conversion by **5%**.
- Set up Metabase dashboards tracking business KPIs such as customer lifetime value, cart abandonment rate and average order value.

Playshifu | playshifu.com | Bangalore, India | 09/2022 – 01/2023
Software Engineer (Full Stack)

- Developed real-time backend services for multiplayer games on **Azure PlayFab**.
- Built internal reporting dashboards in React.js that surfaced **3 major** reporting bottlenecks for the operations team.

Sunstone | sunstone.in | Gurgaon, India | 01/2022 – 08/2022
Software Engineer (Backend)

- Integrated Airtel and Kaleyra cloud calling into the CRM, raising call pickup rates by **35%**.
- Built the CRM backend's REST APIs in Express.js using MVC architecture.

PROJECTS

Gobber | Go | github.com/manojpawarsj12/gobber
An npm-compatible package installer (package.json, package-lock.json v3) with a global store and symlinked node_modules, concurrent HTTP/2 downloads and integrity checks, covered by unit and fake-registry integration tests; benchmarked **~6x faster** than npm on cold installs and **~65x faster** on CI installs.

smolie-js | TypeScript | github.com/manojpawarsj12/smolie-js
A reactive UI library under 1 KB with signals, effects and DOM rendering helpers.

App Tracker | Python, Plotly, PyPI | github.com/manojpawarsj12/apptracker
A desktop time tracker, published on PyPI, that records time spent per app and website and charts it by day, week or month.

EDUCATION

B.E. Computer Science and Engineering | Sathyabama Institute of Science and Technology, Chennai
08/2018 – 07/2022 | CGPA: 8.2/10